## Assignment 3:

Write a C program that includes a user-defined function named isPerfect with the signature int isPerfect(int num);. A perfect number is a positive integer that is equal to the sum of its proper divisors, excluding itself. For example, 28 is a perfect number because the sum of its divisors (1, 2, 4, 7, 14) equals 28.

## Source Code:

```
#include<stdio.h>
int isPerfect(int num) {
   int sum=0,i;
   if (num==0)
      return 0;

   for (i=1;i<=num/2;i++) {
      if (num%i==0) {
         sum=sum+i;
      }
   }
   return sum==num;
}
int main() {
   int n;
   printf("Enter a positive integer number :");
   scanf("%d",&n);
   if (isPerfect(n))
      printf("This is a Perfect number.");
   else
      printf("This is not a Perfect number.");
   return 0;
}
```

## Output:

```
Enter a positive integer number :28

This is a Perfect number.
Process finished with exit code 0
```

```
Enter a positive integer number :12

This is not a Perfect number.
Process finished with exit code 0
```